

CSI ORIGIN 2026

PROBLEM STATEMENT

1 PROBLEM STATEMENT

Hackathon / Event Name : CSI ORIGIN 2026

Problem Statement ID : 8

Title : Autonomous Financial Agent Security

2 BACKGROUND

Current Situation / Context

AI agents are rapidly evolving from systems that provide financial information into autonomous economic actors capable of accessing bank accounts and wallets, initiating transactions, managing portfolios, interacting with DeFi protocols, and making financial decisions on behalf of users and organisations.

This introduces a security model fundamentally different from conventional fintech. Traditional applications generally execute predefined operations under deterministic permissions. An autonomous agent, however, can interpret natural-language instructions, reason over external information, choose actions, and chain multiple actions together. Its behaviour can therefore change dynamically even when its underlying credentials remain valid.

An attacker may therefore not need to steal a private key or directly compromise a financial account. They may instead manipulate the agent's inputs, context, instructions, tools, counterparties, or decision-making process and cause it to voluntarily perform an unauthorised financial action.

Underlying Problem or Need

The core gap is the absence of a security and governance layer specifically designed for autonomous financial agents that can determine whether an intended action is safe, authorised, and consistent with the agent's role before that action reaches real financial infrastructure.

The system must evaluate not only whether an agent possesses valid credentials, but also whether the intended action is consistent with its identity, authority, purpose, behaviour, context, and applicable financial policies.

Why This Is Becoming Important

As autonomous agents gain greater control over financial accounts, wallets, portfolios, and financial protocols, the potential consequences of agent manipulation or compromise increase significantly.

The attack surface also expands beyond conventional credential theft to include prompt and instruction manipulation, malicious external information, compromised tools and APIs, transaction chaining, malicious smart contracts, counterparty impersonation, and manipulation of the agent's financial reasoning.

Limitations of the Current Situation

Conventional financial security mechanisms primarily focus on authentication, credentials, permissions, transaction signatures, and predefined access controls.

These mechanisms can establish that an action is technically authorised or correctly signed, but they may not determine whether the action reflects legitimate agent behaviour or whether the agent was manipulated into initiating it.

At the same time, requiring human approval for every autonomous financial action would significantly reduce the scalability and usefulness of autonomous agents.

Impact of Leaving This Unresolved

Without an appropriate security and governance layer, autonomous financial agents could become high-impact attack surfaces with direct access to financial assets.

A manipulated or compromised agent could voluntarily execute unauthorised transactions, transfer funds to fraudulent counterparties, interact with malicious protocols, or initiate cascading financial actions while remaining technically authenticated.

Conversely, overly restrictive controls could prevent agents from operating autonomously at meaningful scale.

3 PROBLEM

What Is Going Wrong / Missing?

Existing financial security systems do not adequately address the distinction between technical transaction validity and legitimate agent intent.

An autonomous agent may possess valid credentials and generate a correctly signed transaction while the underlying action is the result of manipulated instructions, malicious context, compromised tools, abnormal behaviour, or an unintended chain of decisions.

The missing capability is a security layer that can understand what the agent is trying to do, why it is doing it, whether it is authorised to do so, what risks surround the action, and whether the action should be executed, constrained, delayed, blocked, or escalated.

Who or What Is Affected?

The problem affects autonomous AI agents that access financial accounts, wallets, portfolios, payment systems, DeFi protocols, and other financial infrastructure.

It also affects users, organisations, financial institutions, and asset-management systems that delegate financial authority to autonomous agents.

Why Is It Difficult or Important?

The system must secure autonomous financial agents without eliminating their autonomy.

An agent's behaviour can change dynamically because of new instructions, external information, tool outputs, counterparties, or evolving context. Consequently, static permissions alone cannot fully describe whether a particular action is safe.

The security layer must evaluate identity, authority, intent, behaviour, context, and potential consequences while making decisions quickly enough to remain compatible with autonomous financial workflows.

Consequences

If this problem remains unresolved, autonomous agents may either receive excessive financial authority or become dependent on human approval for routine decisions.

Excessive authority increases the potential impact of compromised or manipulated agents, while excessive human intervention prevents autonomous financial systems from operating efficiently and at scale.

4 CHALLENGE

Design and build a security and governance layer for autonomous financial agents that evaluates an agent's intended financial actions before they reach real financial infrastructure and determines whether those actions should be executed autonomously, constrained, delayed, blocked, or escalated.

The system should establish a continuous understanding of the agent's identity, authority, intent, context, behaviour, counterparties, and potential risks, while continuously monitoring the agent after execution for emerging threats or behavioural deviations.

The System Should Operate as a Continuous Security Loop

Observe → Understand Intent → Verify Authority → Assess Context → Evaluate Risk → Enforce Policy → Execute / Block / Escalate → Monitor → Adapt

Problem Requirements / Constraints

1. The system must establish the identity and applicable authority of the autonomous financial agent before evaluating its intended financial action.
2. The system must interpret the agent's intended action and evaluate whether it is consistent with the agent's authorised role, objectives, and financial policies.
3. The system must evaluate contextual risk factors including transaction characteristics, counterparties, transaction history, external information, tools or APIs involved, and smart-contract or protocol interactions.
4. The system must account for potential manipulation of prompts, instructions, external data, tool outputs, or other inputs that could influence the agent's financial decision-making.
5. The system must detect abnormal or unexpected agent behaviour, including unusual transaction amounts, frequencies, destinations, sequences, or cascading actions.
6. The system must enforce explicit financial controls such as spending limits, permissions, exposure constraints, approved counterparties, and other applicable policies.
7. The system must support graduated autonomy based on assessed risk, allowing low-risk actions to proceed autonomously while subjecting moderate- and high-risk actions to increasingly stronger controls.
8. The system must continuously monitor the agent and executed actions to identify emerging threats, behavioural deviations, compromised workflows, or cascading financial activity.

5 ANNEXURE — DETAILED RULES / CONSTRAINTS

Agent Identity and Authority

The system must determine whether the agent is authorised to perform the intended action and whether the action falls within the financial responsibilities and permissions assigned to it.

Intent Verification

Security evaluation must consider the intended purpose and context of an action rather than relying exclusively on authentication or transaction-signature verification.

A correctly authenticated transaction must not automatically be considered legitimate.

Contextual Risk

The system should evaluate relevant context surrounding a financial action, including transaction history, counterparties, market or financial conditions, tools used, external information, and protocol interactions.

Adversarial Inputs

The system must account for the possibility that the agent's decision-making process has been influenced by malicious instructions, manipulated external content, compromised tools, or other adversarial inputs.

Behavioural Anomalies

The system should identify material deviations from established or expected agent behaviour, including unusual transaction sizes, frequencies, destinations, sequences, or patterns of activity.

Policy Enforcement

The security layer must enforce predefined controls governing financial authority, including spending limits, exposure limits, permitted actions, approved counterparties, and other relevant constraints.

Graduated Autonomy

The system must not treat all actions identically. The level of autonomy granted to the agent should depend on assessed risk:

Low Risk → Execute Autonomously

Moderate Risk → Constrain / Delay / Additional Verification

High Risk → Block / Escalate

Post-Execution Monitoring

Security evaluation must continue after an action has been approved or executed. The system should monitor subsequent behaviour and transaction sequences for signs of compromise, cascading activity, or emerging threats.

6 FINAL PROBLEM CONSTRAINTS

- The system must evaluate intent and context, not merely authentication, credentials, or transaction signatures.
- The system must establish whether an action is consistent with the agent's identity, authority, role, objectives, and financial policies.
- The system must account for prompt and instruction manipulation, malicious external information, compromised tools, malicious protocols, fraudulent counterparties, and abnormal transaction behaviour.
- The system must enforce explicit financial permissions, spending limits, exposure constraints, and other applicable policies.
- The system must provide graduated autonomy, allowing low-risk actions to execute autonomously while applying stronger controls to increasingly risky actions.
- The system must continuously monitor agent behaviour and financial activity after execution to identify emerging threats and cascading actions.
- The system must balance financial security with operational autonomy and must not require human approval for every routine transaction.
- The challenge is focused on establishing Risk-Aware Financial Autonomy, not merely conventional authentication, wallet security, or transaction-level fraud detection.